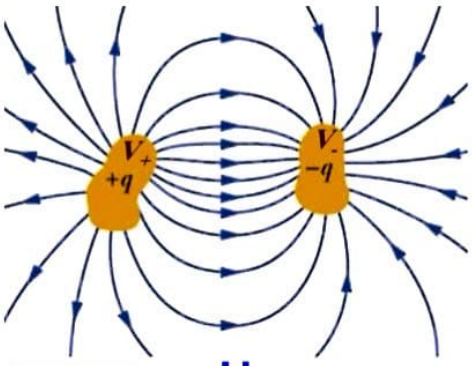
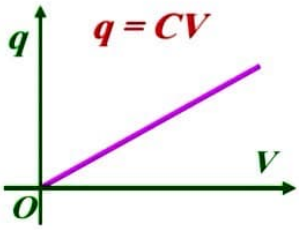


Capacitance

A system of two isolated conductors, one with a charge $+q$ and the other $-q$, separated by an insulator (this can be vacuum or air) is known as a "capacitor." The symbol used to indicate a capacitor is two parallel lines. We refer to the conductors as "plates." We refer to the "charge" of the capacitor as the absolute value of the charge on either plate.

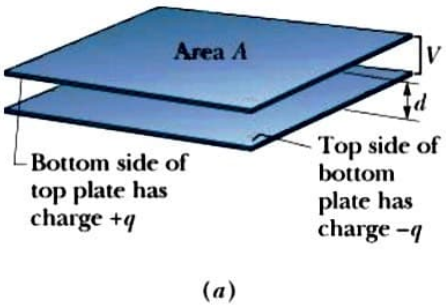
$$C = \frac{q}{V}$$

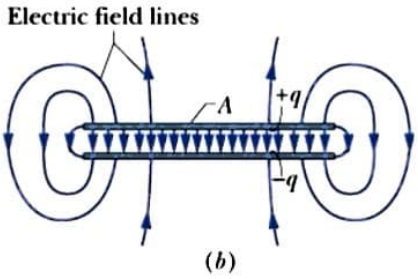


As shown in the figure, the charges on the capacitor plates create an electric field in the surrounding space. The electric potential of the positive and negative plate are V_+ and V_-, respectively. We use the symbol V for the potential difference $V_+ - V_-$ between the plates (ΔV would be more appropriate).

If we plot the charge q as a function of V we get the straight line shown in the figure. The capacitance C is defined as the ratio q/V.

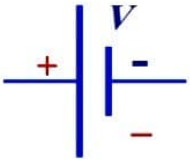
SI Unit : Farad (symbol F) We define a capacitor of $C = 1$ F as one that acquires a charge $q = 1$ C if we apply a voltage difference $V = 1$ V between its plates.





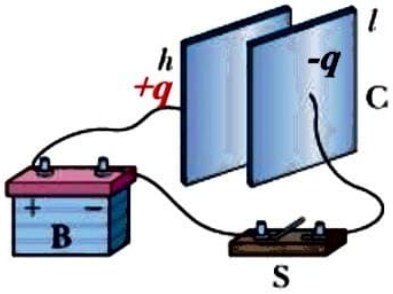
Parallel Plate Capacitor

A parallel plate capacitor is defined as a capacitor made up from two parallel plane plates of area A separated by a distance d. The electric field between the plates and away from the plate edges is uniform. Close to the plates' edges the electric field (known as "fringing field") becomes nonuniform.

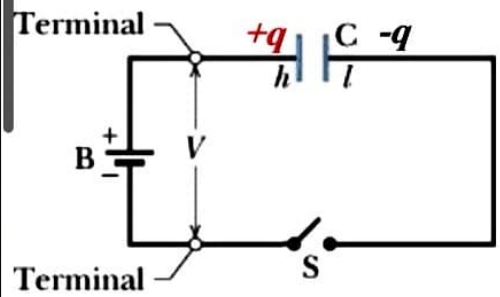


Batteries

A battery is a device that maintains a constant potential difference V between its two terminals. These are indicated in the battery symbol using two parallel lines unequal in length. The longer line indicates the terminal at higher potential while the shorter line denotes the lower-potential terminal.



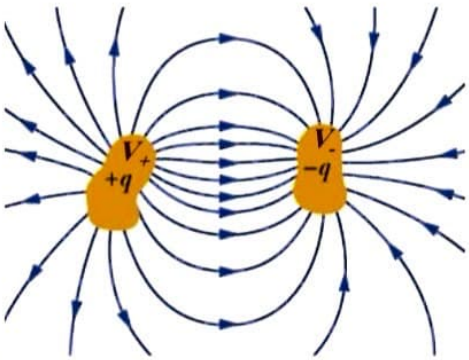
(a)



(b)

Charging a Capacitor

One method to charge a capacitor is shown in the figure. When the switch S is closed, the electric field of the battery drives electrons from the battery negative terminal to the capacitor plate connected to it (labeled "ℓ" for low). The battery positive terminal removes an equal number of electrons from the plate connected to it (labeled "h" for high). Initially the potential difference V between the capacitor plates is zero. The charge on the plates as well as the potential difference between the plates increase, and the charge movement from the battery terminals to and from the plates decreases. All charge movement stops when the potential difference between the plates becomes equal to the potential difference between the battery terminals.



Calculating the Capacitance

The capacitance depends on the geometry of the plates (shape, size, and relative position of one with respect to the other). Below we give a procedure for calculating C.

Recipe :

1. Assume that the plates have charges $+q$ and $-q$.

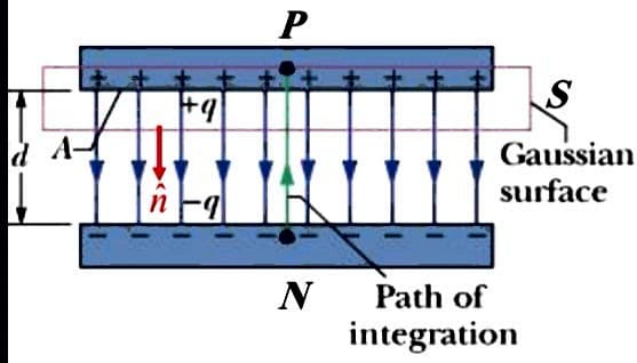
2. Use Gauss' law to determine the electric field $\vec{E}$ between the plates $\left(\varepsilon_0 \oint \vec{E} \cdot d\vec{A} = q_{\text{enc}}\right)$.

3. Determine the potential difference V between the plates using the equation

$V = \int_{-}^{+} \vec{E} \cdot d\vec{s}$ along any path that connects the negative with the positive plate.

4. The capacitance C is given by the equation

$$C = \frac{q}{V}.$$



Capacitance of a Parallel Plate Capacitor

The plates in the figure have area A and are separated by a distance d. The upper plate has a charge $+q$ and the lower plate a charge $-q$.

We apply Gauss' law using the Gaussian surface S shown in the figure.

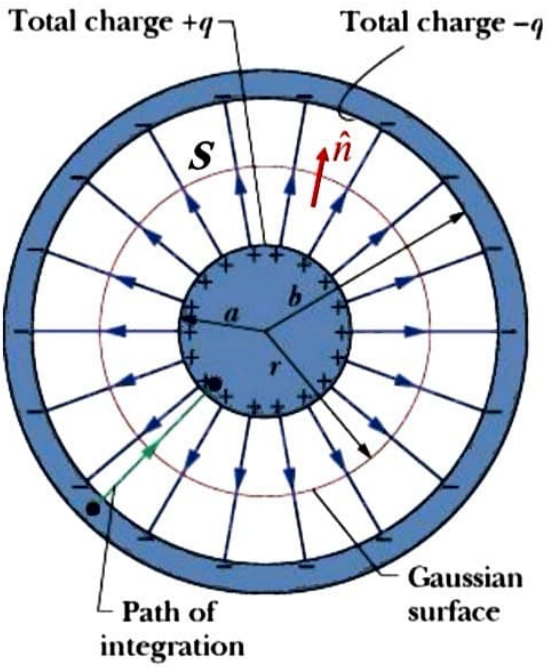
The electric flux $\Phi = EA\cos 0 = EA$.

From Gauss' law we have: $\Phi = \dfrac{q}{\varepsilon_0} \rightarrow EA = \dfrac{q}{\varepsilon_0} \rightarrow E = \dfrac{q}{A\varepsilon_0}$.

The potential difference V between the positive and the negative plate is given by: $V = \int_{-}^{+} Eds\cos 0 = E\int_{-}^{+} ds = Ed = \dfrac{qd}{A\varepsilon_0}$.

The capacitance $C = \dfrac{q}{V} = \dfrac{q}{qd/A\varepsilon_0} = \dfrac{A\varepsilon_0}{d}$.

$$\boxed{C = \frac{A\varepsilon_0}{d}} \qquad \textbf{(25-6)}$$



Cylindrical Capacitor

It consists of two cylinders of radii a and b with a common axis. The two cylinders have a height L. We choose a Gaussian surface S that is also a cylinder with radius r and height L. The flux of the electric field through S is $\Phi = 2\pi rLE\cos 0 = 2\pi rLE$.

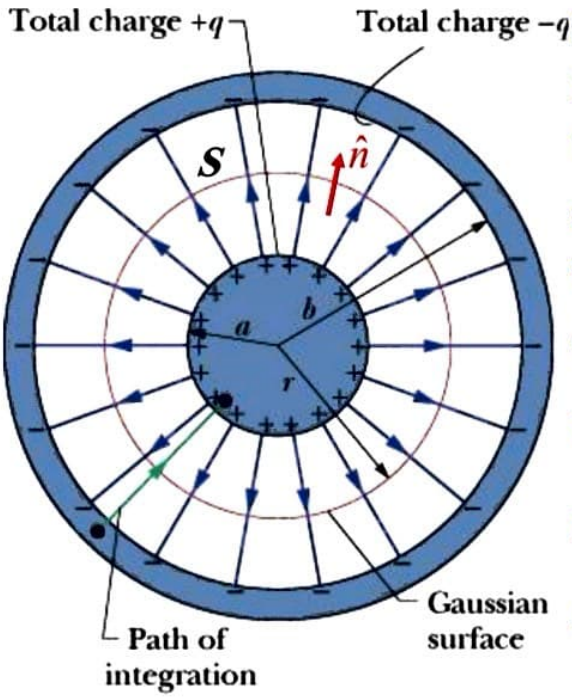
Using Gauss' law we have: $\Phi = \dfrac{q}{\varepsilon_0}$.

If we combine these equations we have:

$$E = \frac{q}{\varepsilon_0 2\pi rL}.$$

The potential difference V between the positive and the negative plate is given by: $V = \int_{-}^{+} Edr\cos 180 = -\dfrac{q}{\varepsilon_0 2\pi L}\int_b^a \dfrac{dr}{r} = -\dfrac{q}{\varepsilon_0 2\pi L}[\ln r]_b^a = \dfrac{q}{\varepsilon_0 2\pi L}\ln\left(\dfrac{b}{a}\right)$.

The capacitance $C = \dfrac{q}{V} = \dfrac{q}{(q/2\pi L\varepsilon_0)\ln(b/a)} = \dfrac{2\pi L\varepsilon_0}{\ln(b/a)}$. **(25-7)**



Spherical Capacitor

It consists of two concentric spheres of radii a and b. We choose a Gaussian surface S that is also a sphere with radius r.

The flux of the electric field through S is:

$$\Phi = 4\pi r^2 E\cos 0 = 4\pi r^2 E.$$

Using Gauss' law we have: $\Phi = \dfrac{q}{\varepsilon_0}$.

If we combine these equations we have:

$$E = \frac{q}{4\pi\varepsilon_0 r^2}.$$

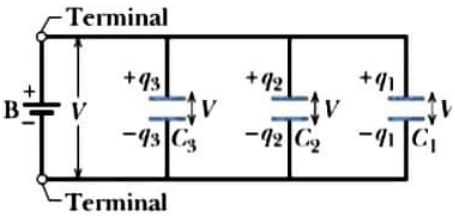
The potential difference V between the positive and the negative plate is given by:

$$V = \int_-^+ E dr \cos 180 = -\frac{q}{4\pi\varepsilon_0}\int_b^a \frac{dr}{r^2} = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r}\right]_b^a = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right).$$

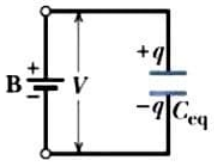
The capacitance

$$C = \frac{q}{V} = \frac{q}{\dfrac{q}{4\pi\varepsilon_0}\left(\dfrac{1}{a}-\dfrac{1}{b}\right)} = \frac{4\pi\varepsilon_0}{\left(\dfrac{1}{a}-\dfrac{1}{b}\right)} = 4\pi\varepsilon_0\left(\frac{ab}{b-a}\right).$$

(25-8)



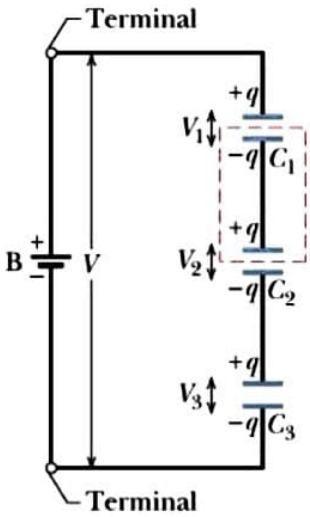
(a)



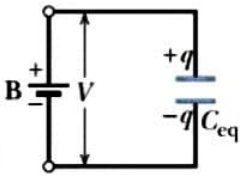
(b)

Equivalent Capacitor

Consider the combination of capacitors shown in the figure to the left and to the right (upper part). We will substitute these combinations of capacitor with a single capacitor C_{eq} that is "electrically equivalent" to the capacitor group it substitutes.



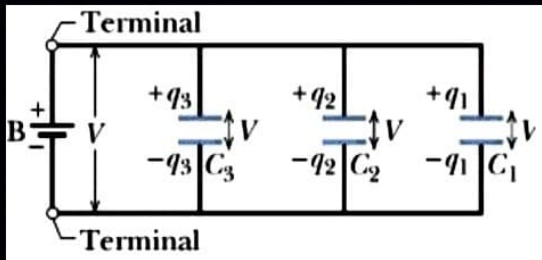
(a)



(b)

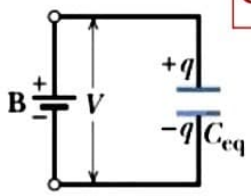
This means that if we apply the same voltage V across the capacitors in fig. a and fig. b (either right or left) by connecting to a battery, the same charge q is provided by the battery. Alternatively, if we place the same charge q on plates of the capacitors in fig. a and fig. b (either right or left), the voltage V across them is identical. This can be stated in the following manner: If we place the capacitor combination and the equivalent capacitor in separate black boxes, by doing electrical measurements we cannot distinguish between the two.

(25-9)



(a)

$$C_{eq} = C_1 + C_2 + C_3$$



(b)

Capacitors in Parallel

In fig. *a* we show three capacitors connected in parallel. This means that the plate of each capacitor is connected to the terminals of a battery of voltage V. We will substitute the parallel combination of fig. *a* with a single equivalent capacitor shown in fig. *b*, which is also connected to an identical battery.

The three capacitors have the **same potential difference** V across their plates.

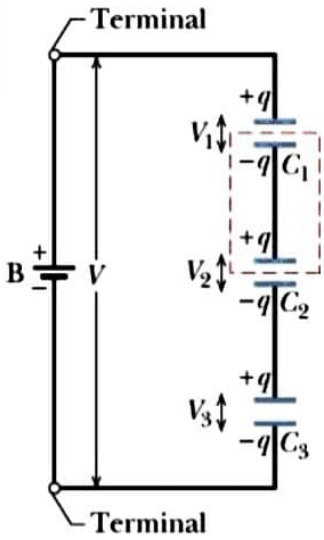
The charge on C_1 is $q_1 = C_1 V$. The charge on C_2 is $q_2 = C_2 V$.

The charge on C_3 is $q_3 = C_3 V$. The net charge $q = q_1 + q_2 + q_3 = (C_1 + C_2 + C_3)V$.

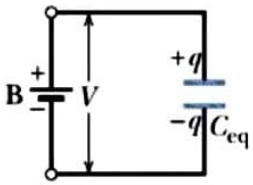
The equivalent capacitance $C_{eq} = \dfrac{q}{V} = \dfrac{(C_1 + C_2 + C_3)V}{V} = C_1 + C_2 + C_3$.

For a parallel combination of n capacitors it is given by the expression:

$$C_{eq} = C_1 + C_2 + \ldots + C_n = \sum_{j=1}^{n} C_j \qquad \textbf{(25-10)}$$



(a)



(b)

Capacitors in Series

In fig. *a* we show three capacitors connected in series. This means that one capacitor is connected after the other. The combination is connected to the terminals of a battery of voltage V. We will substitute the series combination of fig. *a* with a single equivalent capacitor shown in fig. *b*, which is also connected to an identical battery.

The three capacitors have the **same charge** q on their plates.

The voltage across C_1 is $V_1 = q / C_1$.

The voltage across C_2 is $V_2 = q / C_2$.

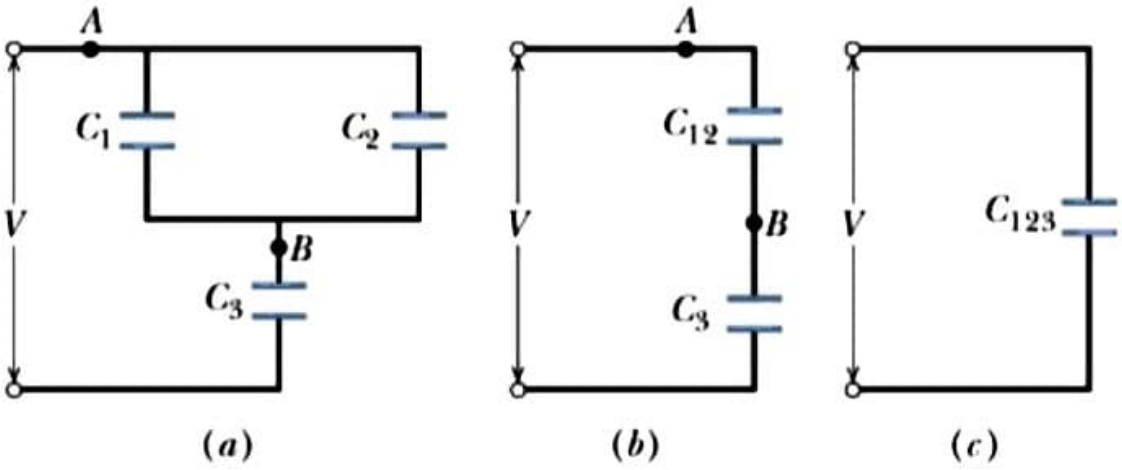
The voltage across C_3 is $V_3 = q / C_3$.

The net voltage across the combination $V = V_1 + V_2 + V_3$.

Thus we have: $V = q\left(\dfrac{1}{C_1} + \dfrac{1}{C_2} + \dfrac{1}{C_3}\right)$.

The equivalent capacitance $C_{eq} = \dfrac{q}{V} = \dfrac{q}{q\left(\dfrac{1}{C_1} + \dfrac{1}{C_2} + \dfrac{1}{C_3}\right)} \rightarrow$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \qquad \textbf{(25-11)}$$

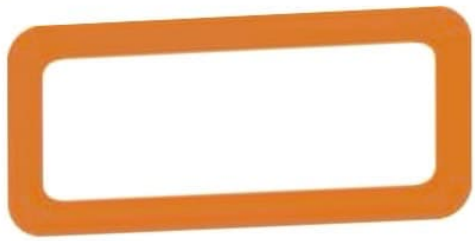


More Complex Capacitor Systems

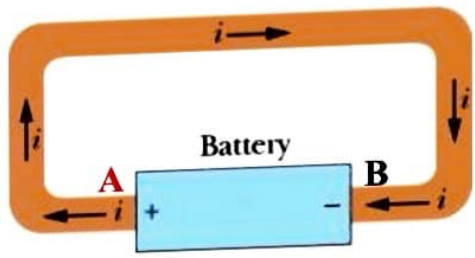
In general, a capacitor system may consist of smaller capacitor groups that can be identified as connected "in parallel" or "in series."

For example, C_1 and C_2 in fig. *a* are connected in parallel. They can be substituted by the equivalent capacitor $C_{12} = C_1 + C_2$ as shown in fig. *b*. Capacitors C_{12} and C_3 in fig. *b* are connected in series. They can be substituted by a single capacitor C_{123} as shown in fig. *c*.

C_{123} is given by the equation $\dfrac{1}{C_{123}} = \dfrac{1}{C_{12}} + \dfrac{1}{C_3}$.



(a)

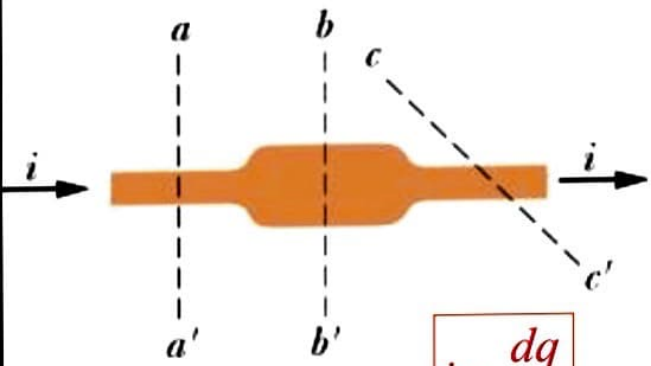


(b)

Electric Current

Consider the conductor shown in fig. *a*. All the points inside the conductor and on its surface are at the same potential. The free electrons inside the conductor move in random directions and thus there is no net charge transport.

We now make a break in the conductor and insert a battery as shown in fig. *b*. Points *A* and *B* are now at potentials V_A and V_B, respectively ($V_A - V_B = V$, the voltage of the battery). The situation is not static any more, but charges move inside the conductor so that there is a net charge flow in a particular direction. We define this net flow of electric charge as "electric current."



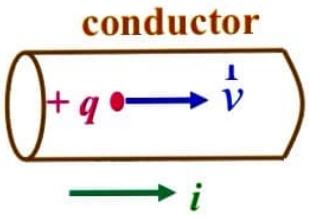
$$i = \frac{dq}{dt}$$

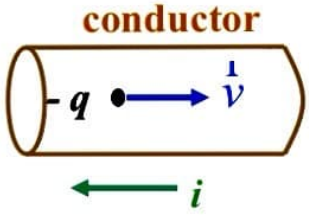
Consider the conductor shown in the figure. It is connected to a battery (not shown) and thus charges move through the conductor. Consider one of the cross sections through the conductor (aa' or bb' or cc').

The electric current i is defined as $i = \dfrac{dq}{dt}$.

Current = rate at which charge flows

Current SI Unit: C/s, known as the "ampere"



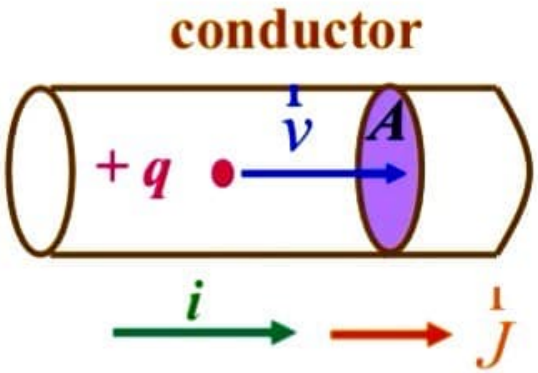


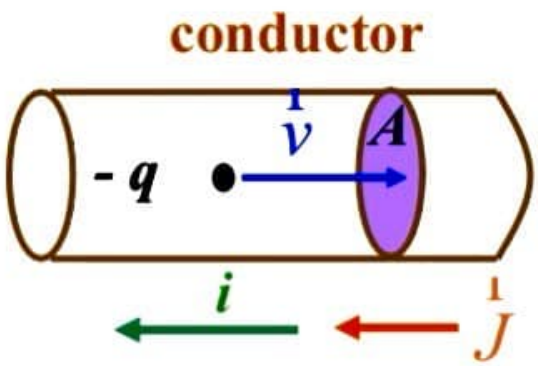
Current Direction

An electric current is represented by an arrow, which has the same direction as the charge velocity. The sense of the current arrow is defined as follows:

1. If the current is due to the motion of **positive** charges, the current arrow is **parallel** to the charge velocity $\vec{v}$.

2. If the current is due to the motion of **negative** charges, the current arrow is **antiparallel** to the charge velocity $\vec{v}$.





Current Density

Current density is a vector that is defined as follows:

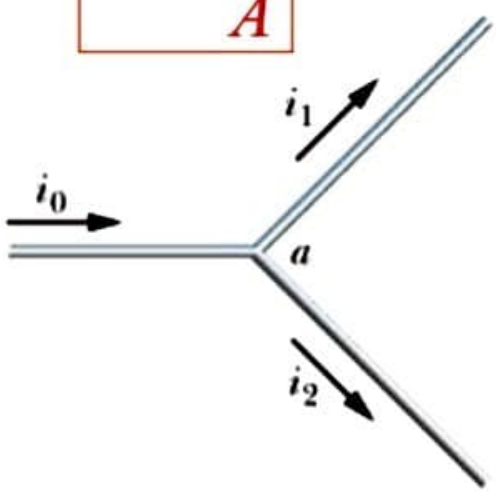
Its magnitude is $J = \dfrac{i}{A}$ **SI unit for J: A / m²**

The direction of $\vec{J}$ is the same as that of the current.

The current through a conductor of cross-sectional area A is given by the equation $i = JA$ if the current density is constant.

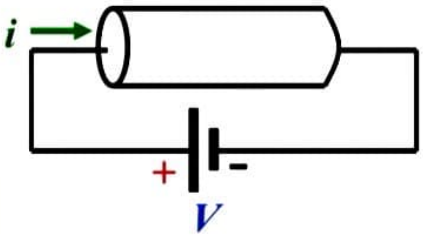
If $\vec{J}$ is not constant, then $i = \int \vec{J} \cdot d\vec{A}$.

$$\boxed{J = \frac{i}{A}}$$



We note that even though the current density is a vector the electric current is not. This is illustrated in the figure to the left. An incoming current i_0 branches at point a into two currents, i_1 and i_2.

Current $i_0 = i_1 + i_2$. This equation expresses the conservation of charge at point a. Please note that we have not used vector addition.



$$R = \frac{V}{i}$$

Resistance

If we apply a voltage V across a conductor (see figure) a current i will flow through the conductor.

We define the conductor resistance as the ratio $R = \frac{V}{i}$.

SI Unit for R: $\frac{V}{A} =$ the ohm (symbol Ω)

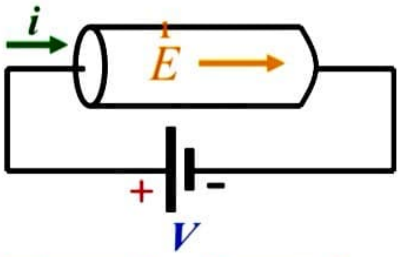
A conductor across which we apply a voltage $V = 1$ volt and results in a current $i = 1$ ampere is defined as having resistance of 1 Ω.

Q: Why not use the symbol "O" instead of "Ω"?

A: Suppose we had a 1000 Ω resistor.

We would then write: 1000 O, which can easily be mistakenly read as 10000 Ω.

A conductor whose function is to provide a specified resistance is known as a "resistor." The symbol is given to the left.



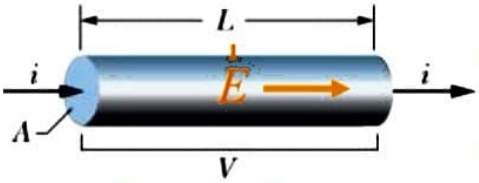
$$\vec{E} = \rho \vec{J} \qquad \vec{J} = \sigma \vec{E}$$

Resistivity

Unlike the electrostatic case, the electric field in the conductor of the figure is not zero. We define as resistivity ρ of the conductor the ratio $\rho = \frac{E}{J}$.

In vector form: $\vec{E} = \rho \vec{J}$.

SI unit for ρ: $\frac{\text{V/m}}{\text{A/m}^2} = \frac{\text{V}}{\text{A}}\text{m} = \Omega \cdot \text{m}$



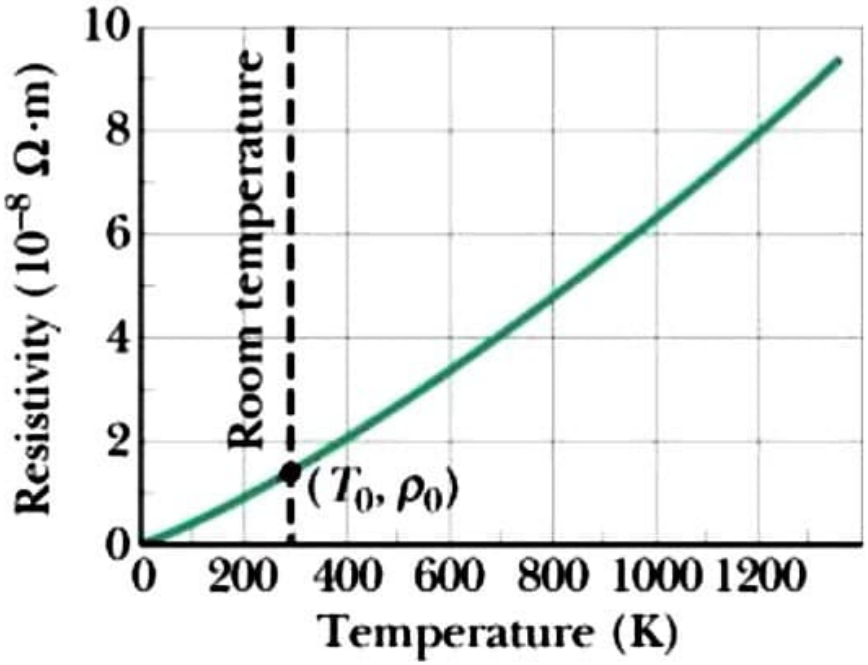
$$R = \rho \frac{L}{A}$$

The conductivity σ is defined as $\sigma = \frac{1}{\rho}$.

Using ρ, the previous equation takes the form: $\vec{J} = \sigma \vec{E}$.

Consider the conductor shown in the figure above. The electric field inside the conductor is $E = \frac{V}{L}$. The current density is $J = \frac{i}{A}$. We substitute E and J into equation $\rho = \frac{E}{J}$ and get: $\rho = \frac{V/L}{i/A} = \frac{V}{i}\frac{A}{L} = R\frac{A}{L} \rightarrow R = \rho\frac{L}{A}$.

$$\rho - \rho_0 = \upsilon_0 \alpha \left(T - T_0\right)$$



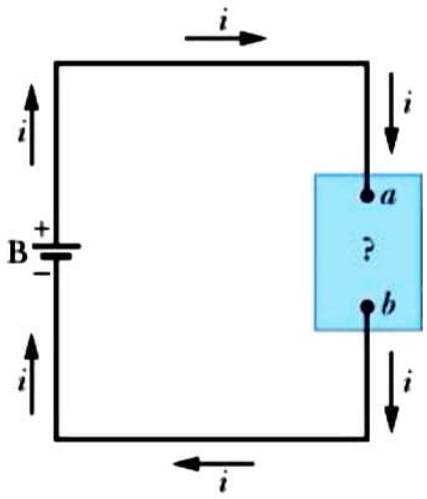
Variation of Resistivity with Temperature

In the figure we plot the resistivity ρ of copper as a function of temperature T. The dependence of ρ on T is almost linear. Similar dependence is observed in many conductors.

The following empirical equation is used for many practical applications: $\rho - \rho_0 = \upsilon_0 \alpha \left(T - T_0\right)$. The constant α is known as the "temperature coefficient of resistivity." The constant T_0 is a reference temperature usually taken to be room temperature ($T_0 = 293$ K), and ρ_0 is the resistivity at T_0. For copper, $\rho_o = 1.69 \times 10^{-8}\ \Omega \cdot \text{m}$.

Note : Temperature enters the equation above as a difference $\left(T - T_0\right)$. Thus either the Celsius or the Kelvin temperature scale can be used.

$$\boxed{P = iV}$$

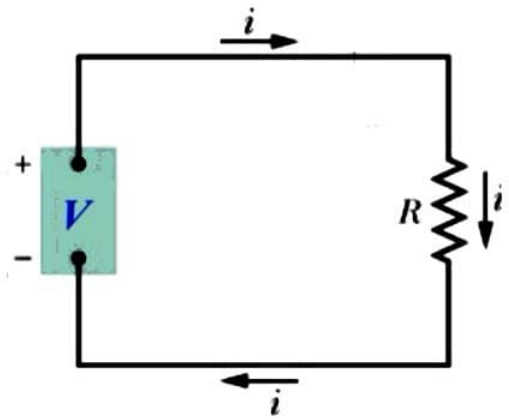


Power in Electric Circuits

Consider the circuit shown in the figure. A battery of voltage V is connected across the terminals a and b of a device. This can be a resistor, a motor, etc. The battery maintains a potential difference V between the terminals a and b and thus a current i flows in the circuit as shown in the figure. During the time interval dt a charge $dq = idt$ moves between the terminals. We note that $V_a > V_b$.

The potential energy of the charge decreases by an amount $dU = Vdq = Vidt$. Using energy conservation we conclude that the lost energy has been transferred by the battery to the device and has been converted into some other form of energy. The rate at which energy is transferred to the device is known as "**power**" and it is equal to $P = \frac{dU}{dt} = \frac{Vidt}{dt} = Vi$.

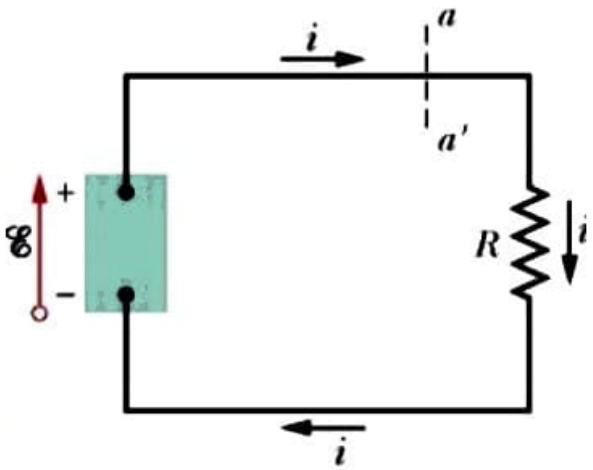
SI unit for P : $\mathrm{V \cdot A}$ It is known as the "watt" (symbol W).



$$\boxed{P = i^2 R} \qquad \boxed{P = \frac{V^2}{R}}$$

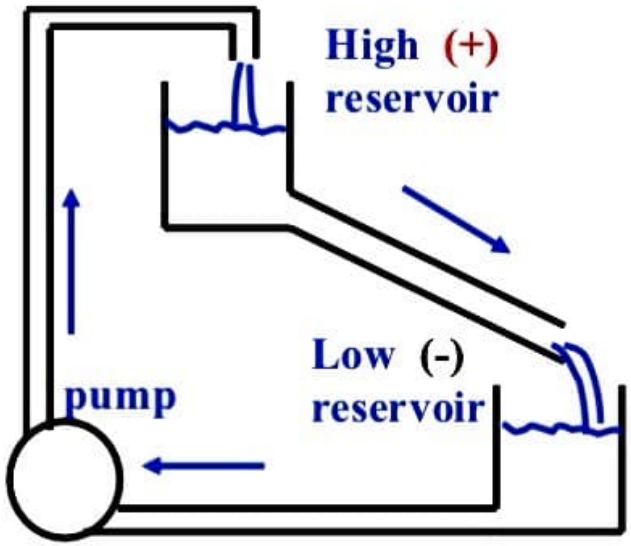
If the device connected to the battery is a resistor R then the energy transferred by the battery is converted as **heat** that appears on R. If we combine the equation $P = iV$ with Ohm's law $i = \frac{V}{R}$, we get the following two equivalent expressions for the rate at which heat is dissipated on R:

$$P = i^2 R \qquad \text{and} \qquad P = \frac{V^2}{R}$$



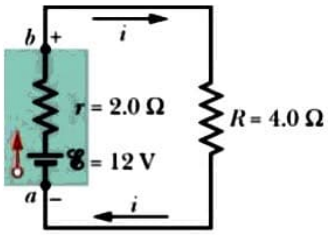
In order to create a current through a resistor, a potential difference must be created across its terminals. One way of doing this is to connect the resistor to a battery. A device that can maintain a potential difference between two terminals is called a **"seat of an emf "** or an **"emf device."** Here emf stands for electromotive force. Examples of emf devices are a battery, an electric generator, a solar cell, a fuel cell, etc.

These devices act like "charge pumps" in the sense that they move positive charges from the low-potential (negative) terminal to the high-potential (positive) terminal. A mechanical analog is given in the figure below.



In this mechanical analog a water pump transfers water from the low to the high reservoir. The water returns from the high to the low reservoir through a pipe, which is the analog of the resistor.

The emf (symbol E) is defined as the potential difference between the terminals of the emf device when no current flows through it.

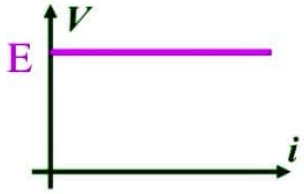


Ideal and Real Emf Devices

An emf device is said to be **ideal** if the voltage V across its terminals a and b does **not** depend on the current i that flows through the emf device: $V = \mathrm{E}$.

$V = \mathrm{E}$

Ideal emf device

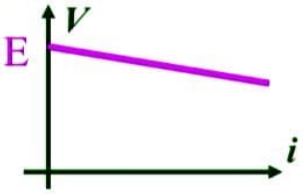


An emf device is said to be **real** if the voltage V across its terminals a and b **decreases** with current i according to the equation $V = \mathrm{E} - ir$.

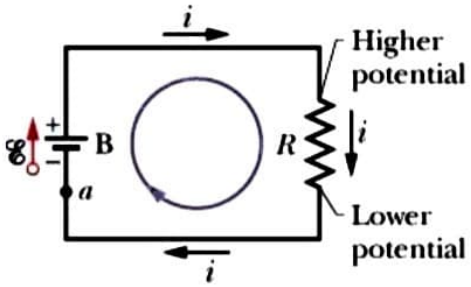
The parameter r is known as the "**internal resistance**" of the emf device.

$V = \mathrm{E} - ir$

Real emf device



$\mathrm{E}i - iR = 0$



Current in a Single - Loop Circuit

Consider the circuit shown in the figure. We assume that the emf device is ideal and that the connecting wires have negligible resistance. A current i flows through the circuit in the clockwise direction.

In a time interval dt a charge $dq = idt$ passes through the circuit. The battery is doing work $dW = \mathrm{E}dq = \mathrm{E}idt$.i Using energy conservation we can set this amount of work equal to the rate at which heat is generated on R: $\mathrm{E}idt = Ri^2dt \rightarrow$

$\mathrm{E}i = Ri \rightarrow \mathrm{E}i - iR = 0.$

Kirchhoff put the equation above in the form of a rule known as Kirchhoff's loop rule (KLR for short).

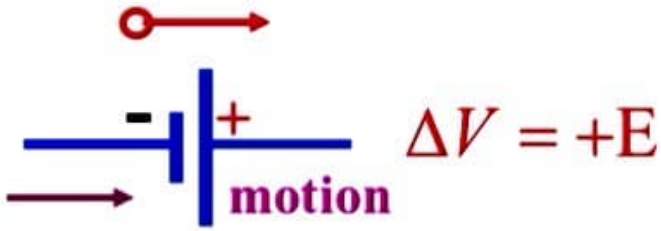
KLR : The algebraic sum of the changes in potential encountered in a complete traversal of any loop in a circuit is equal to zero.

The rules that give us the algebraic sign of the charges in potential through a resistor and a battery are given on the next page.

Resistance Rule :

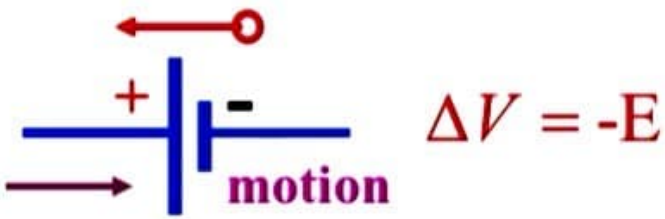
For a move through a resistance in the direction of the current, the change in the potential is $\Delta V = -iR$.

For a move through a resistance in the direction opposite to that of the current, the change in the potential is $\Delta V = +iR$.

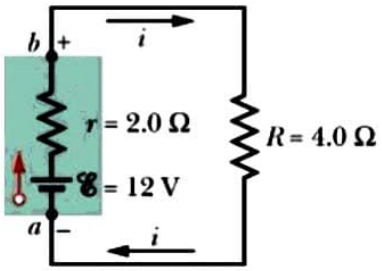


EMF Rule :

For a move through an ideal emf device in the direction of the emf arrow, the change in the potential is $\Delta V = +E$.



For a move through an ideal emf device in a direction opposite to that of the emf arrow, the change in the potential is $\Delta V = -E$..



Potential Difference Between Two Points :

Consider the circuit shown in the figure. We wish to calculate the potential difference $V_b - V_a$ between point b and point a.

$V_b - V_a =$ sum of all potential changes ΔV along the path from point a to point b.

We choose a path in the loop that takes us from the initial point a to the final point b.

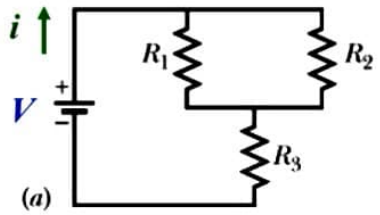
$V_f - V_i =$ sum of all potential changes ΔV along the path.

There are two possible paths: We will try them both.

Left path: $V_b - V_a = \mathrm{E} - ir$

Right path: $V_b - V_a = iR$

Note : The values of $V_b - V_a$ we get from the two paths are the same.

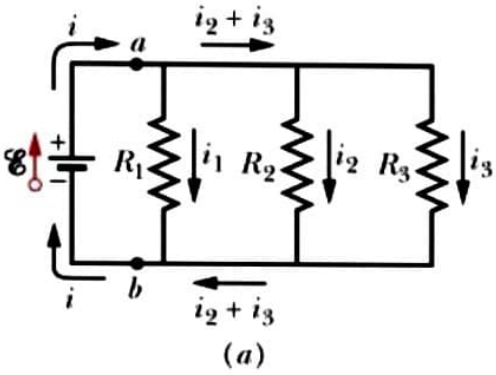


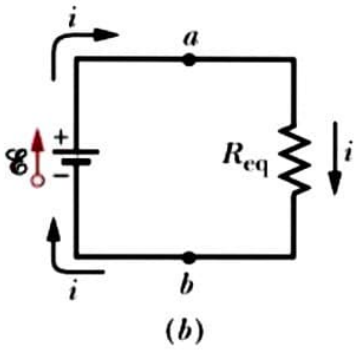
Equivalent Resistance

Consider the combination of resistors shown in the figure. We can substitute this combination of resistors with a single resistor R_{eq} that is "electrically equivalent" to the resistor group it substitutes.

This means that if we apply the same voltage V across the resistors in fig. a and across R_{eq}, the same current i is provided by the battery. Alternatively, if we pass the same current i through the circuit in fig. a and through the equivalent resistance R_{eq}, the voltage V across them is identical. This can be stated in the following manner: If we place the resistor combination and the equivalent resistor in separate black boxes, by doing electrical measurements we cannot distinguish between the two.

Resistors in Parallel





$$\boxed{\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Consider the three resistors shown in the figure. "In parallel" means that the terminals of the resistors are connected together on both sides. Thus resistors in parallel have the **same potential** applied across them. In our circuit this potential is equal to the emf E of the battery. The three resistors have different currents flowing through them. The total current is the sum of the individual currents. We apply KJR at point *a*:

$$i = i_1 + i_2 + i_3 \qquad i_1 = \frac{E}{R_1} \qquad i_2 = \frac{E}{R_2} \qquad i_3 = \frac{E}{R_3} \rightarrow$$

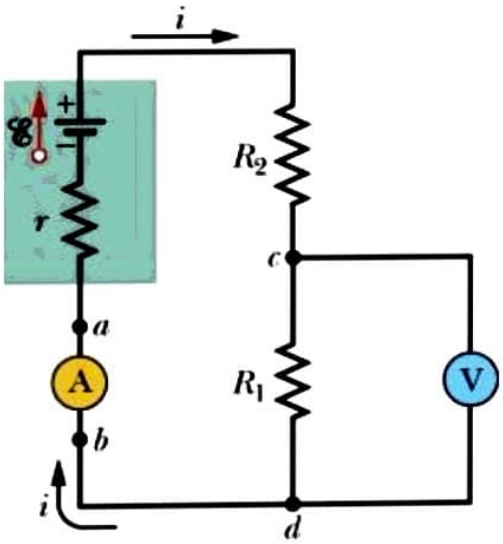
$$i = E\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)$$ (eq. 1). From fig. *b* we have:

$$i = \frac{E}{R_{eq}}$$ (eq. 2). If we compare equations 1 and 2 we get:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}.$$

(27-13)

Ammeters and Voltmeters



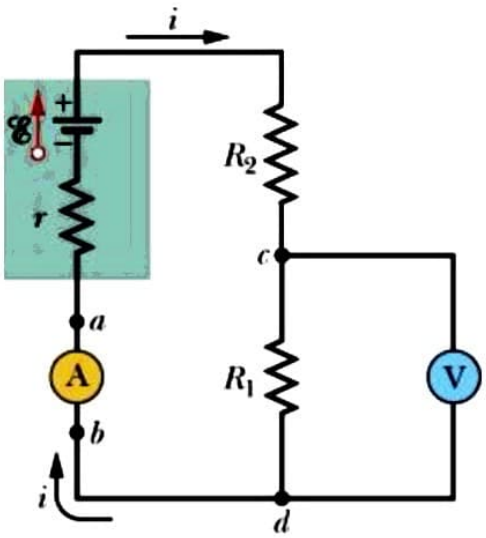
An ammeter is an instrument that measures current. In order to measure the current that flows through a conductor at a certain point we must cut the conductor at this point and connect the two ends of the conductor to the ammeter terminals so that the current can pass through the ammeter. An example is shown in the figure, where the ammeter has been inserted between points *a* and *b*.

It is essential that the ammeter resistance R_A be much smaller than the other resistors in the circuit. In our example: $R_A = R_1$ and $R_A = R_2$.

A voltmeter is an instrument that measures the potential difference between two points in a circuit. In the example of the figure we use a voltmeter to measure the potential across R_1. The voltmeter terminals are connected to the two points *c* and *d*.

It is essential that the voltmeter resistance R_V be much larger than the other resistors in the circuit. In our example: R_V ? R_1 and R_V ? R_2.

(27-14)



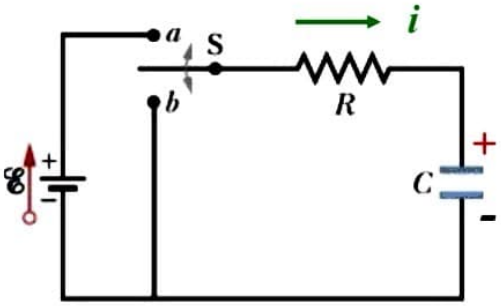
Ammeters and Voltmeters

An ammeter is an instrument that measures current. In order to measure the current that flows through a conductor at a certain point we must cut the conductor at this point and connect the two ends of the conductor to the ammeter terminals so that the current can pass through the ammeter. An example is shown in the figure, where the ammeter has been inserted between points a and b.

It is essential that the ammeter resistance R_A be much smaller than the other resistors in the circuit. In our example: $R_A = R_1$ and $R_A = R_2$.

A voltmeter is an instrument that measures the potential difference between two points in a circuit. In the example of the figure we use a voltmeter to measure the potential across R_1. The voltmeter terminals are connected to the two points c and d.

It is essential that the voltmeter resistance R_V be much larger than the other resistors in the circuit. In our example: R_V ? R_1 and R_V ? R_2.



RC Circuits : Charging of a Capacitor

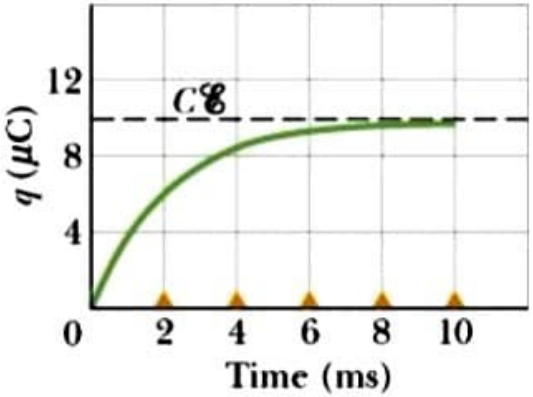
Consider the circuit shown in the figure. We assume that the capacitor is initially uncharged and that at $t = 0$ we throw the switch S from the middle position to position a. The battery will charge the capacitor C through the resistor R.

Our objective is to examine the charging process as a function of time.

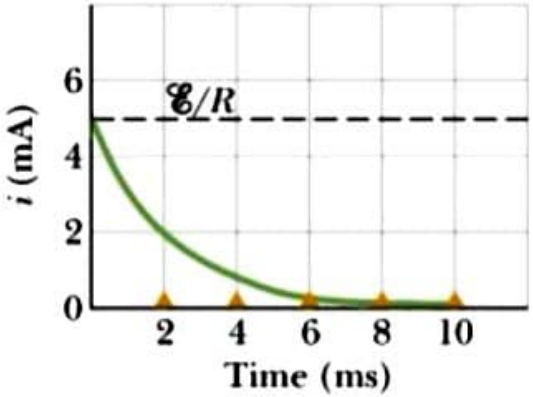
We will write KLR starting at point b and going in the counterclockwise direction:

$\mathrm{E} - iR - \frac{q}{C} = 0$. The current $i = \frac{dq}{dt} \rightarrow \mathrm{E} - \frac{dq}{dt}R - \frac{q}{C} = 0$. If we rearrange the terms we have: $\frac{dq}{dt}R + \frac{q}{C} = \mathrm{E}$. This is an inhomogeneous, first order, linear differential equation with initial condition $q(0) = 0$. This condition expresses the fact that at $t = 0$ the capacitor is uncharged.

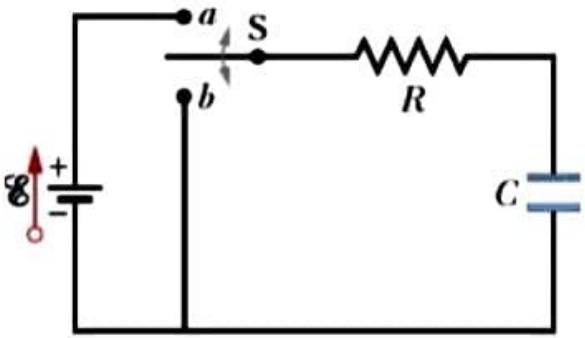
$$\boxed{\tau = RC}$$



(*a*)



(*b*)



Differential equation: $\dfrac{dq}{dt}R + \dfrac{q}{C} = \mathrm{E}$

Intitial condition: $q(0) = 0$

Solution: $q = C\mathrm{E}\left(1 - e^{-t/\tau}\right)$ Here: $\tau = RC$

The constant τ is known as the "time constant" of the circuit. If we plot q versus t we see that q does not reach its terminal value CE but instead increases from its initial value and reaches the terminal value at $t = \infty$. Do we have to wait for an eternity to charge the capacitor? In practice, no.

$q(t = \tau) = (0.632)C\mathrm{E}$

$q(t = 3\tau) = (0.950)C\mathrm{E}$

$q(t = 5\tau) = (0.993)C\mathrm{E}$

If we wait only a few time constants the charge, for all practical purposes, has reached its terminal value CE.

The current $i = \dfrac{dq}{dt} = \left(\dfrac{\mathrm{E}}{R}\right)e^{-t/\tau}$. If we plot i versus t we get a decaying exponential (see fig. *b*).